DATA 505 (Statistics Using R): Exam 1

Part 2: Open Computer

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Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are "not alive" (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed `.qmd` file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze data about Antarctic penguins from the Palmer Station research program. The data are available in the `palmerpenguins` R package.

```r
# Install and load the required packages
if(!"palmerpenguins" %in% installed.packages()){
  install.packages("palmerpenguins")
}
if(!"ggplot2" %in% installed.packages()){
  install.packages("ggplot2")
}

library(palmerpenguins)
library(ggplot2)

# Load the penguins data
```

```r
data("penguins")

library(dplyr)
```

Data Exploration (4 points)

1 (2 points)

Use the `str()` function to examine the structure of the `penguins` dataset.

```r
# YOUR CODE HERE
str(penguins)
```

```
tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
 $ species          : Factor w/ 3 levels "Adelie","Chinstrap",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ island           : Factor w/ 3 levels "Biscoe","Dream",..: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm   : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm    : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g      : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex              : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year             : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

How many observations and variables are there?

Answer: 344 observations and 8 variable WRITE YOUR ANSWER HERE.

2 (2 points)

Extract the `bill_length_mm` column from the penguins dataset and store it in a new variable called `bill_lengths`. Then use the `summary()` function to examine this variable.

```r
# YOUR CODE HERE
bill_lengths <- penguins$bill_length_mm
summary(bill_lengths)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
  32.10   39.23   44.45   43.92   48.50   59.60       2 
```

Research Question

We want to investigate whether the mean bill length of Adelie penguins differs significantly from 39 mm. This research question is motivated by previous studies suggesting that 39 mm might be a typical bill length for this species.

1 (4 points)

Filter the dataset to include only Adelie penguins and store the result in a new variable called `adelie_penguins`. Remove any rows with missing bill length measurements.

```r
# YOUR CODE HERE
adelie_penguins <- penguins %>%
  filter(species == "Adelie") %>%
  filter(!is.na(bill_length_mm))
adelie_penguins
```

```
# A tibble: 151 x 8
   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
 1 Adelie  Torgersen           39.1          18.7               181        3750
 2 Adelie  Torgersen           39.5          17.4               186        3800
 3 Adelie  Torgersen           40.3          18                 195        3250
 4 Adelie  Torgersen           36.7          19.3               193        3450
 5 Adelie  Torgersen           39.3          20.6               190        3650
 6 Adelie  Torgersen           38.9          17.8               181        3625
 7 Adelie  Torgersen           39.2          19.6               195        4675
 8 Adelie  Torgersen           34.1          18.1               193        3475
 9 Adelie  Torgersen           42            20.2               190        4250
10 Adelie  Torgersen           37.8          17.1               186        3300
# i 141 more rows
# i 2 more variables: sex <fct>, year <int>
```

2 (3 points)

Extract the bill lengths from the Adelie penguin data and store them in a variable called `adelie_bills`. How many Adelie penguins have complete bill length measurements?

```r
# YOUR CODE HERE
adelie_bills <- adelie_penguins$bill_length_mm
str(adelie_bills)
```

```
 num [1:151] 39.1 39.5 40.3 36.7 39.3 38.9 39.2 34.1 42 37.8 ...
```

**Number of Adelie penguins with complete data:*151* Write your answer here.

3 (5 points)

Calculate the following descriptive statistics for Adelie penguin bill lengths: - Mean - Standard deviation
- Median - 25th and 75th percentiles

```r
# YOUR CODE HERE
summary(adelie_bills)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  32.10   36.75   38.80   38.79   40.75   46.00 
```

```r
sd(adelie_bills)
```

```
[1] 2.663405
```

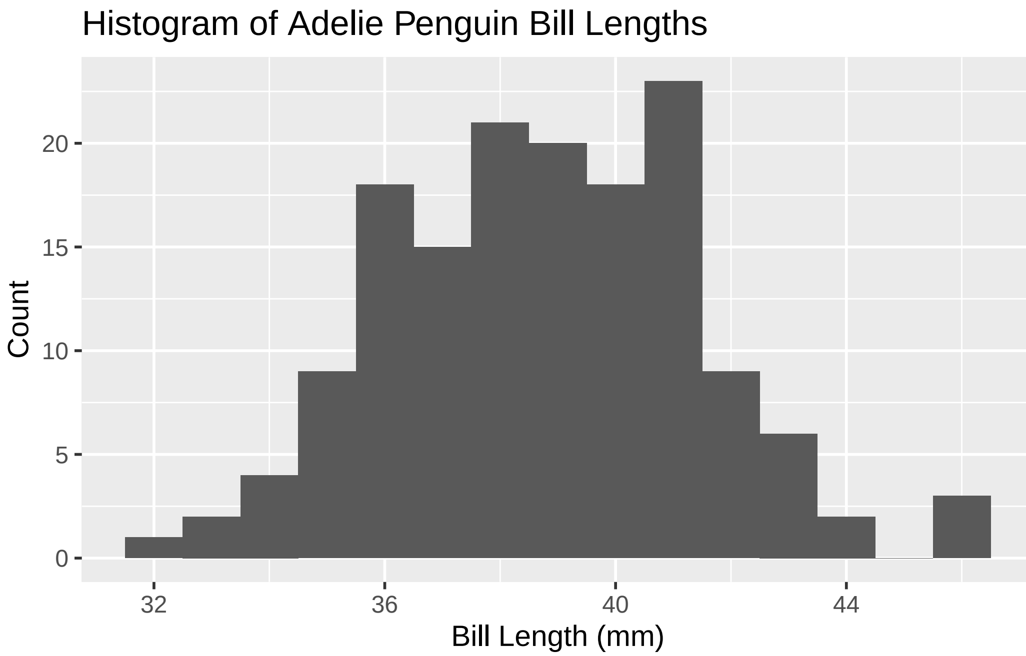
```r
#- Mean: 38.79mm
#- Standard deviation :2.663405
#- Median:38.80
#- 25th percentiles: 36.75
#- 75th percentiles: 40.75
```

4 (5 points)

Create a histogram of Adelie penguin bill lengths using ggplot2 or base R. Include appropriate axis labels and a title.

```r
# YOUR CODE HERE
library(ggplot2)

ggplot(data = adelie_penguins, aes(x = bill_length_mm)) +
  geom_histogram(binwidth = 1) +
  labs(
    title = "Histogram of Adelie Penguin Bill Lengths",
    x = "Bill Length (mm)",
    y = "Count"
  )
```



Based on the histogram, briefly describe the distribution of bill lengths (address shape, center, and spread):

Description:the distribution of Bill Length is a symmetrical distribution because median and mean mainly the same. . The center of the distribution is median which the median is 38.80. the spread is 36.75 to 40.75 Write your answer here.

Statistical Analysis

1 (5 points)

Now we will formally test whether the population mean bill length μ differs from 39 inches, i.e. we will test:

$$H_0 : \mu = 39$$

against

$$H_A : \mu \neq 39$$

Perform a two-sided t-test to test these hypotheses. Use $\alpha = 0.05$.

```r
# YOUR CODE HERE
t_test_result <- t.test(adelie_bills, mu = 39, alternative = "two.sided")
t_test_result
```

```
    One Sample t-test

data:  adelie_bills
t = -0.96246, df = 150, p-value = 0.3374
alternative hypothesis: true mean is not equal to 39
95 percent confidence interval:
 38.36312 39.21966
sample estimates:
mean of x
 38.79139
```

Based on your test results, what is your conclusion? Interpret the results in the context of the research question.

Conclusion: p-value is more that alpha (0.3374 > 0.05).fail to reject H_0. This means that the sample does not provide strong evidence that the mean bill length is different from 39 mm. The two-sided t-test shows that the mean bill length of Adelie penguins is 38.79 mm, with a 95% confidence interval of 38.36 to 39.22 mm. In short, The mean bill length of Adelie penguins is 38.79 mm, very close to 39 mm. The p-value is high, so any difference is likely due to random variation rather than a real difference in the population. Write your conclusion here.

2 (4 points)

Define another variable (a numeric vector) `confidence_interval` and assign as its value the two bounds (lower and upper) of a 95% confidence interval for the mean bill length of Adelie penguins. Print the result. Full points are possible only if you do not "hard code" the bounds for the interval, i.e. if you create this vector directly by using R functions applied to the dataset. Half points maximum if you define the vector by typing the entries manually.

```r
# YOUR CODE HERE
confidence_interval <-t.test(adelie_bills)$conf.int
confidence_interval
```

```
[1] 38.36312 39.21966
attr(,"conf.level")
[1] 0.95
```